Testing BLDC motor's Phase Wiring - Hall Sensors and Wiring.

Tommycat
Giga Member

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

Testing BLDC motor's Phase Wiring - Hall Sensors and Wiring.

#1

07-19-2020, 09:27 AM

Testing electrical components of a typical hub motor with a digital multi meter.

First take a few minuets and look over the wiring going into the motor for cuts, scrapes, missing insulation and other wiring issues or poor connections that should be resolved first. Smell for evidence of overheating or burning that could have damaged the winding insulation, wires, or connections.

Two major electrical systems need to be checked out in a typical BLDC hub motor that has hall sensors.

1) The motor's 3 phase windings. (high voltage components-thicker wires)
2) The motor's 3 hall sensors, (low voltage components-thinner wires) if present. If not, you will need a "sensor less" capable controller and skip these tests.

Motor Winding Testing.

- With the 3 phase wire leads (heavy gauge wires) disconnected from everything and not moving or turning the motor. Test resistance (ohms)between two phases at a time and comparing all three phase combination's resistances. All three readings should match. Note that this is just a ***general*** check with a Digital Multi Meter. As the resistance readings are very low to be real accurate. It's not so much an ***exact*** same matching reading, but if they all are ***comparable***. Very easily can be much less than 1 ohm.
- Then check resistance between each phase wire at a time and a metal part of the motor. It should read infinity or open circuit. Nothing shorted or any resistance reading to ground. If so, do not use the motor until repairs (if possible) are made.
- Then tie two phase wires together in different combinations at a time for a total of three tests. At each combination, a similar **magnetic resistance** (also known a "cogging ') should be felt while trying to turn the wheel by hand. With no resistance present when all wires are separated.

Note: If resistance to turning by hand is felt when the motor is connected to the controller, and NOT with the motor disconnected from the controller. This would indicate bad FETs, I.E problem with the controller. See an excellent tutorial on testing controller MOSFETS here... GRIN: Testing for Blown Mosfets
Also on a geared hub motor, wheel will have to be turned in reverse to engage the motor.

- And lastly, if your able to spin your motor with a drill or other set-up at the ***same constant speed*** for all testing. Spin the motor up and first check for hot spots in the windings. Then compare the ***AC voltage outputs*** of all three combinations of phase wires. They should all be close to the same.

Note: A slight degradation of the winding insulation may not show up in a resistance test. So, by putting some load, or voltage stress on the windings as in this last test. Will give a more accurate assessment of the health of the motor. By all means, use a LCR meter if available.

Motor Hall Sensor Testing.

Always check the wiring and connections first for any damage or poor connection.

Overview...

In a BLDC motor that has hall sensors. The hall sensors are used to determine the actual position of the rotor for accurate sensored type controller operation. Typically, they are mounted on a PCB board, with the sensors close to and facing the motor's magnets. Or they can be mounted at the correct positions directly in or besides the motor's stators, facing the magnets. (see hall sensor picture below for sensing surface) Note: Depending on the motor's phase angle, and the separation between each sensor used by the manufacturer. It would not be unusual to have the center hall sensor face, facing the opposite direction as the other two. In addition, it could be a different model that would invert its individual output. I.E. They all won't necessarily have to be facing the same direction or be the same part. Label each sensor's location and orientation before removal.

The operational characteristics below describe a typical SS41F, bipolar, latching, motor hall sensor often used. Which will give a good over-all view. But be aware that there are other flavors of motor hall sensors that perform somewhat differently. I.E. Unipolar or Omnipolar type. Always identify your hall sensor's part number and look up its data sheet for particulars before testing or replacement. I.E. Bench testing requirements change for these different types of sensors.

Each individual hall sensor has three wires going to it, as it requires a voltage input (typically 5vdc) , 0vdc, (batt negative, or ground), with the last wire being signal. This will bring the wiring harness for the hall sensors to a total of 5. Typically, RED=5vdc+, BLACK=0vdc, and each of the last three are the signal wires of their respective sensor, YELLOW, BLUE, and GREEN.

Note: Some motors may include a 6th wire in this bundle. Typically, white in color and often used to connect to an internal thermal sensor. It may also be used to provide a signal for a speedometer input, having its own hall sensor separate and apart from the motor's hall sensors.

Some motors may provide an extra set of hall sensor wiring and sensors. One is only used at a time, with the other as a back-up or spare.

When effected by the motor's magnet's gauss, ether positive or negative passing by the sensor. It will switch the hall sensor output's signal voltage from HIGH (typically 5 or 3.3vdc as provided by the controller **to** the signal wire) to LOW or 0vdc, or vice versa. It does this by **grounding (or not) the reference electronic signal voltage** supplied by the controller. (Much like how the electronic cruise or low brake signal wires are shorted to ground to trigger their actions.) This electronic digital (On/Off) signal is used by the controller to accurately determine stator to rotor position.

To make this operation perfectly clear... looking at it from a somewhat **electromechanical** point of view for clarity. (hopefully)

The hall sensor uses the 5vdc supply and ground from the controller for **operation only**. It does not use this power in the output or for the signal reference.
This supply voltage just enables the sensor to work, from this point of view, like power that would energize a contactor's coil.
What switches the coil's power off and on, is the change of magnet gauss between the positive and negative magnetic poles. One will latch the coil "On", and the other polarity will latch it "OFF". Note: With "latching" type hall sensor as mentioned above...
The sensor's "contacts" have the **signal voltage** from the **controller** on one side. With the systems **ground** on the other side.
When the coil is energized, the contacts "close", shorting the electronic signal to ground producing a "low" signal state. Conversely, when the coil is de-energized with the change of magnetic polarity. The contacts "open" allowing the electronic signal to go back to its "pull-up" or" high" signal state.

This electronically recognized digital voltage switching, at the correct time, and in the correct order is used by the controller to determine when each of its phase winding FETs are actuated. Causing the motor to move in the correct direction, at the desired speed or current.

One last <u>bottom line</u> for those breezing thru... :-)

Controller **provides** regulated 5vdc and ground to each sensor. (Power)
Controller **provides** an individual 5 to 3.3vdc **electronic pull up voltage** to each hall sensor's signal wire.
(You can test for this signal voltage while having the motor hall sensors disconnected, at the controller's connector.)
Change of magnetic polarity will turn sensor "ON", resulting in electronic signal being shorted to ground. (0vdc to ground)
Change to opposite magnetic polarity will turn sensor "OFF", resulting in signal not being shorted to ground. (5vdc to ground)

For replacement a well-known and often used hall sensor used in BLDC motors to determine rotor position would be the Honeywell SS41 series.
Here is the data sheet for the G model.

https://www.mouser.com/datasheet/2/1...ns-1894281.pdf

1) Testing when connected to a working controller.

See this tutorial...

https://www.ebikes.ca/documents/Hall...stingFinal.pdf

If you find a hall sensor's signal output stuck on 0vdc, check **Note 1:** seen below.

2) Bench testing not using a controller.

As a controller provides a reference signal voltage to the hall sensor's output terminal, typically 5vdc to 3.3vdc. This being the voltage used that is shown switching between the HIGH and LOW values during testing. (5 to 3.3vdc... and 0vdc) This voltage will have to be supplied and added to the circuit when testing the motor by itself. This is done by adding a **10K resistor** connected between the 5vdc input voltage and the signal output of the hall sensor being tested. (see picture below)

The external 5vdc source can be provided by a USB 5vdc type wall wort transformer, 3- 1.5vdc AA batteries in series, or USB type rechargeable cell phone back-up battery which I prefer. Current or amperage requirements are low. (<25mAs typical...) The 3 total should use less than 36 mA current **maximum**.

Note: Geared hub motors need to be turned in reverse, as the clutch will not move the motor components going forward. Direct drive will work in either direction.

Why is the 10K resistor required?

As discussed earlier, the hall sensor is used to short the electronic signal to ground. With-out the controller limiting this voltage during a bench test as the signal is being provided by us. The current must be reduced or throttled to an acceptable level that the hall sensor can handle. For a SS41G the maximum drop-dead output current level is 20mA. So, a 10K resistor at 5vdc will keep it well below this level at about .5mA and has been proven to work well for bench testing. A 1K resistor @ 5vdc will allow 5mA if more of a "stress" test is desired. All the way up to 250 ohms @ 5vdc which will allow 20mA. **The point is you just can't wire it directly!**

See this tutorial...

https://electricbike.com/forum/forum...tor-controller

And this...

Example of testing set-up...

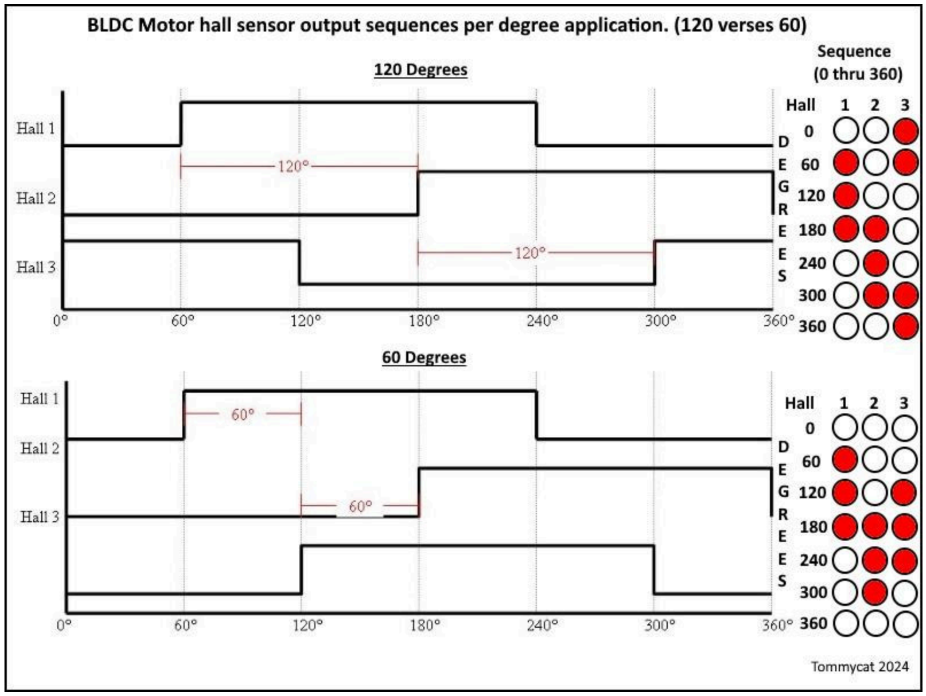
Note 1: If during the non-controller testing, all hall sensor operations check out correctly but your still having an issue. While disconnected from the motor, test for voltage at the **controller's** motor hall sensor input wires. You should read a steady electronic signal voltage provided by the controller of typically 5vdc to 3.3vdc to all 3 hall sensors. If this positive reference voltage is missing, check wiring to the PCB board, or replacement of the controller may be required.

Bad Testing Results?

If properly done initial testing indicates problems, the motor will have to be opened up to gain access to the internals. To further narrow down the cause of the problem between wiring, connections, or parts.

Hall Sensor to Phase Wiring Configuration

After the motor's wiring and components have been verified as good. The motor's phase wiring to hall sensor locations need to be correctly identified and wired for the correct controller output.



If you have a controller with "learning wires" or "auto configuration" by all means put it to use. As the controller will magically configure itself. If not, use this chart to successfully pair the hall sensors to their respective phase windings...

Take it easy on the throttle till better results are obtained! (see Testing Recommendation box in the chart.)

Regards,
T.C.

Last edited by Tommycat; 1 day ago.

See my completed Magic Pie V5 rear hub motor E-Bike build HERE.

Tags: None

Likes 3

Rondak1000
Newbie

Join Date: Jun 2021
Posts: 1
Location: Melbourne, VIC, Australia

06-27-2021, 11:10 PM #2

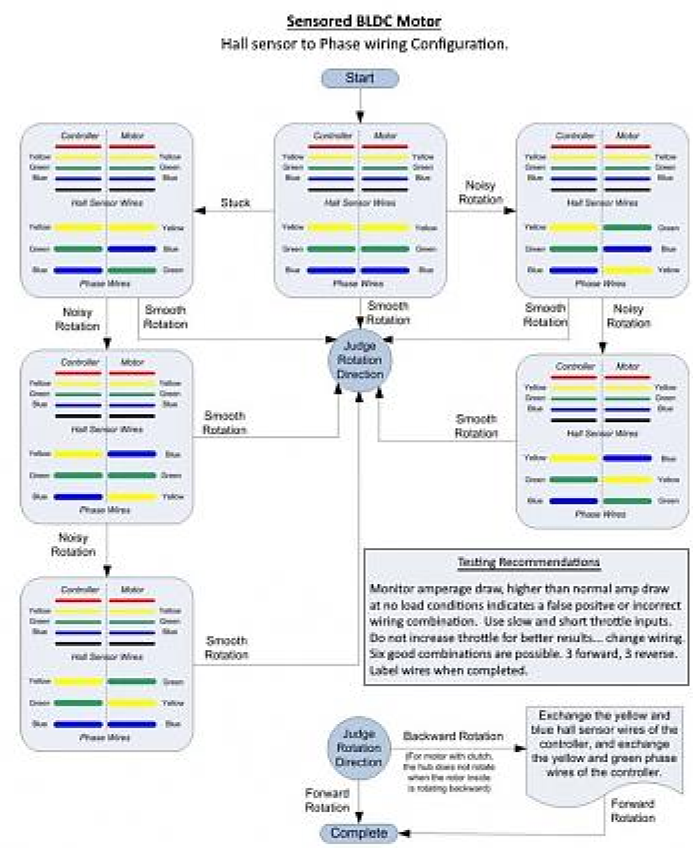
Thanks for this comprehensive write up @Tommycat

Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

10-30-2021, 02:47 AM #3

Hello Tommycat, I love your diagram for BDLC testing! (attached for your reference)



I was wondering, could this could be used for trying to match any controller to a hub motor?

I have a 48V 250W pedal assist bicycle which is working fine.
I bought a KT sine wave controller to replace the original because with the original one I'm stuck at 23km/h and nothing is programmable.

I tried connecting the new controller and it turns on fine with its own new KT display, however:

- **it shows 0% battery charge although I know its fully charged.**
- **nothing works, meaning the motor doesnt spin using the throttle, nor with pedal movement using the pedal assist sensor.**
- **Clearly this controller doesnt "talk" to the motor.**

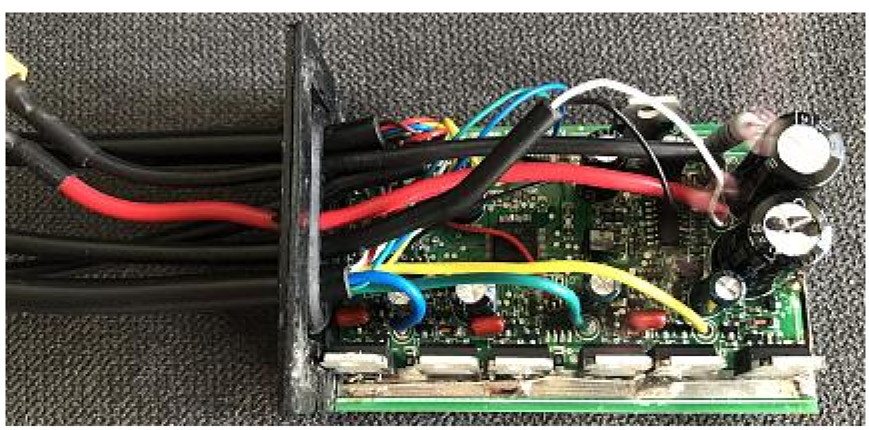
how would you proceed in a similar situation?

- **can I use your diagram to try out those combinations?**
- **but more importantly, how can I do that minimising the chances of blowing out everything? (this battery doesnt have a switch, so once its plugged is immediately on)**

I attach pics of the old and new controllers

OLD

NEW



Thank you in advance for any thought you could share with me.
Greetings from Singapore!
Valerio

Likes 1

Tommycat
Giga Member

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

10-30-2021, 06:30 AM #4

> *Originally posted by **Valerio***
> ***I was wondering, could this could be used for trying to match any controller to a hub motor?***

Yes. Any three phase motor with hall sensors to a compatible controller. Note: some controllers have a "learning" wire or programming that can determine the correct wiring on their own.
From your description, no noise, no growling, no effort to try and turn the motor... this doesn't seem to be the main issue to start with.

> *Originally posted by **Valerio***
> *I tried connecting the new controller and it turns on fine with its own new KT display, however.*

Was the display and controller sold as a ***set***? What model KT display is it? Can you provide the direct link to the kit?

> *Originally posted by **Valerio***
> ***it shows 0% battery charge although I know its fully charged.***

Typical issue with a miss-matched controller/display... see display questions above.

> *Originally posted by **Valerio***
> ***but more importantly, how can I do that minimising the chances of blowing out everything? (this battery doesnt have a switch, so once its plugged is immediately on)***

I thought KT displays incorporated an on/off button. But yes with direct battery power going to the controller possible tips...

Unplug the battery and let the controller's capacitors discharge completely before changing possible motor wiring or disconnecting/reconnecting external devices.

Curious, where does the throttle attach... do you have a proper wiring diagram?
Go thru all the KT display's set-up parameters making sure they are correct.

Regards,
T.C.

See my completed Magic Pie V5 rear hub motor E-Bike build HERE.

Valerio
Newbie

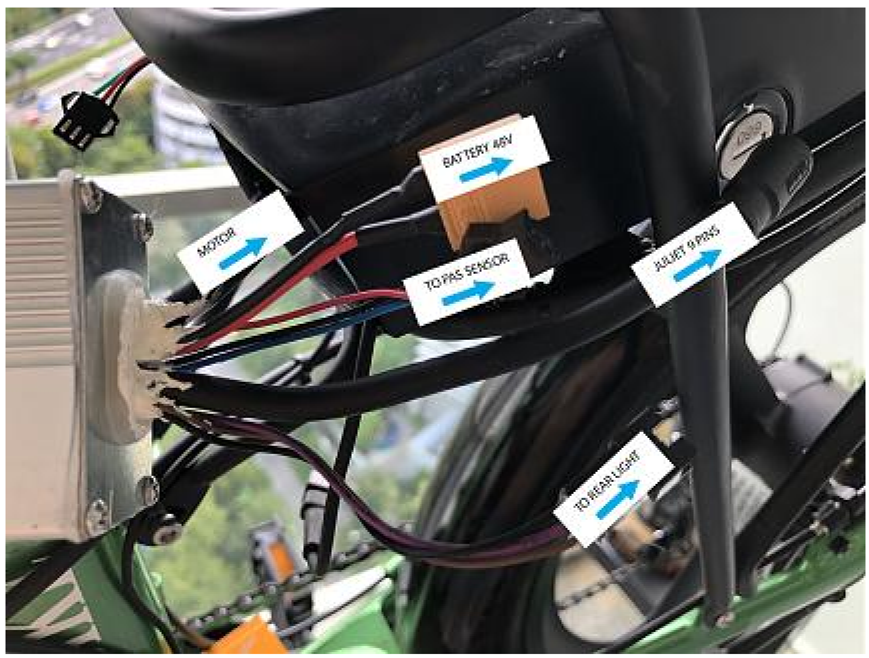
Join Date: Feb 2021
Posts: 16
Location: Singapore

10-30-2021, 09:08 PM #5

Hey Tommycat thanks a lot for your quick reply,

This the original link of the controller (I got the Sinewave 14A)
and this is the display (have to select the KT-LCD5).

Once connected to each other, I can turn on and navigate thru the parameters and reset, change speed etc...
But has I said, nothing else.
The original controller doesnt have any cable to connect a throttle (which is illegal in singapore).
I attache a pic here



- So when I tested the new controller (which has a throttle plug) I used a throttle to see if it could run at all, but nothing.
- I also tried moving the pedal to activate the PAS sensor, still no sign of movement or noise.
- I also tried to hold down the PAS down button on the display to see if could start moving in "walking mode". Nothing.
- Resetted all the parameters in the new display and tried again. Nothing

I read in other posts around that KT controllers and Displays seem to be pretty ok in communicating with each other even when they are not sold as kit. I had a brief messaging exchange with the seller (whom I dont thing

knows much) and I got the LCD manual with all the parameters
which I attach here, but most sounds like sanskrit to me...

[ATTACH]n143780[/ATTACH]

Would it be possible that some of the wiring inside the new controller is not the same of the original one?
if it turns on, clearly the current goes thru properly, but still theres this communication issue.

Do you think its worthy to try re-shuffle these cables between controller and motor (Im talking about the 9pins plug) or it sounds like a lost cause?
Thanks again.

Tommycat
Giga Member

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

10-31-2021, 10:42 AM #6

I like the idea of your trying the "walk" mode feature, although with no positive results... but the more clues the better!

> *Originally posted by **Valerio***
> *Would it be possible that some of the wiring inside the new controller is not the same of the original one?*
> *if it turns on, clearly the current goes thru properly, but still theres this communication issue.*

I would imagine the wiring internal to the controllers would be similar. It's the wire colors and the connections to the outside peripheral components that concern me. There doesn't seem to be much of any set rules on connections and wiring, I.G. wiring colors, ETC. between manufactures.

With this in mind, two issues I think should be addressed first.

1) No communication between the display and the controller. (I don't think I know of a display that talks directly to a motor...)

2) Check for 5vdc controller regulated power to the throttle, PAS wires or motor's hall sensors supply. Which ever is most convenient to access. If available would verify the controller is properly energized by the control wire from the display, and/or not being accidently shorted out by mis-wiring.

My theories...
With loss of communication, the display shows a battery with no voltage. So the display thinks that the battery voltage is too low, and stops any motor operation because of the Low Voltage Cut-Out setting. (40 + or - .05vdc as seen on the controller's sticker.)
No communication can also be caused by not having any 5vdc controller regulated voltage output. (see #2 above and check for voltage.)

I would recommend disconnecting every thing except for the battery, and display.

Turn on and check for error codes. And then for 5vdc output.
If all is well, plug in motor and re-check. Then throttle. Looking to see if at any time communication begins, and when it stops.
Look for and write down any error codes.

> *Originally posted by **Valerio***
>
> *Do you think its worthy to try re-shuffle these cables between controller and motor (Im talking about the 9pins plug) or it sounds like a lost cause?*

Hopefully just unplugging at this time will show a link to the issue. Right now I'm more suspicious of how the 8 wire main and connected wiring loom are connected to the throttle for one.
Remember, ***NEVER trust wire colors to be correct or the same.***

Way to early for a lost cause! ;-)

See my completed Magic Pie V5 rear hub motor E-Bike build HERE.

Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

10-31-2021, 07:17 PM #7

Hey Tommycat,

to your points:

> *1) No communication between the display and the controller. (I don't think I know of a display that talks directly to a motor...)*

I respectfully disagree :) I think the KT new controller and his display talk to each other without issue, I can even navigate thru the parameters..
perhaps you meant the issue is btw CONTROLLER and MOTOR?

> *2) Check for 5vdc controller regulated power to the throttle, PAS wires or motor's hall sensors supply. Which ever is most convenient to access. If available would verify the controller is properly energized by the control wire from the display, and/or not being accidently shorted out by mis-wiring.*

I think I need a bit of guidance here, whats is the **5vdc controller regulated power?**
would you be able to simplify this whole point? or point me in the right direction? I do have a tester but I think I only know how to use for basic things...

after I understood properly your second point I can move forward.

btw we'll have to start thinking about a way for me to return this favour to you!
V.

11-01-2021, 06:33 AM #8

Tommycat
Giga Member

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

The parameters of the display are stored there and will write it to the controller as well as retrieve current setpoints back ... I'm still skeptical... Can you turn your headlight on? Can you make any of the status displays function. (I.E. throttle twist?)

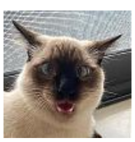
See my completed Magic Pie V5 rear hub motor E-Bike build HERE.

Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

11-01-2021, 06:40 AM #9

I can't turn the headlight on, but I can enter the modification parameters mode, like speed limit, while size etc. as well as P and C parameters...



Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

11-02-2021, 05:26 AM #10

wouldnt this be the proof that the broken link is btw Controller and Motor?

Likes 1

Tommycat
Giga Member

11-02-2021, 06:38 AM #11

Perhaps we are not on the same page as regards to "communication".

When all is well, the display "communicates" with it's controller by electronic digital means. Not just a matter of a correct analog voltage on the correct wires. This "communication" depends on a few things.

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

1) 5vdc regulated power provided by an energized controller.
2) Correct wiring between the display and controller. Which includes Data 1, Data 2, and Ground wire.
3) A functioning display and controller using the same communication protocols. (I.G. TTL) And the same programming.

Display functions such as the battery display, and ability to turn on your lights depend on this communication. (as well as everything else that makes it work...) This includes the ability to set the controller's operational setpoints by way of the display's software and programming parameters.

Because you can see the display's side working, doesn't automatically mean the controller is receiving this information and acting on it. If you can get information FROM the controller to be displayed on the display's panel, then I would be some what satisfied... The controller is capable of setting a throttle or motor hall sensor error. My thought was if you had unplugged these devices or inputs to the controller (as mention previously), it may trigger an error which would mean communication is established. It also should show other communicated statuses such as BRAKE, THROTTLE, LIGHTS, and SPEED.

As seen in my diagram above, motor wiring and function are base electrical with no electronic "communication" present. Can there be an issue with connection. Absolutely! Another reason I requested you try it with the motor disconnected. But at this time it sounds like the controller is not even trying to make it work, putting this on the back burner.

Which brings me circling back to item #1 above.

"Check for 5vdc controller regulated power to the throttle, PAS wires or motor's hall sensors supply. Which ever is most convenient to access. If available would verify the controller is properly energized by the control wire from the display, and/or not being accidently shorted out by mis-wiring."

Your display takes full battery power on terminal #1 RED wire, and when the display is turned on. Sends full battery power to terminal #2, down the BLUE wire to the controller. Since you have a functioning display, we know that the GROUND wire, terminal #3, BLACK wire is good.
When the controller receives the full battery input from the BLUE wire. It energizes it's own internal regulated 5vdc power supply, providing power to all of it's electronic components as well as power for the throttle, PAS and motor's hall sensors (supply and sensor signals)

A quick check would be at the connector for the throttle. *TYPICALLY* 5vdc should be present between the controller's RED and BLACK throttle wires. Check for power like you would to see if a 9 volt battery is good. This should determine where to go next...

For more information on testing your hall sensor throttle, and to determine the exact wiring for example... see this thread.

[Guide to Hall Sensor Throttle operation, testing, and ...](https://electricbike.com/forum/forum/kits/golden-mo...)
https://electricbike.com/forum/forum/kits/golden-mo...

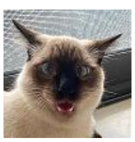
***** 8-16-2018 Welcome to the Hall Sensor Throttle Thread! If there is something you'd like to add, correct, needs better explanation, or have a question about......

When checking full E-bike battery power be very careful! As it has a lot of power potential!!!

For more specific and accurate checkpoint suggestions. A detailed wiring map, or up close and labeled pictures of your wiring harness and connections would be most helpful.

Please clarify, in your original controller's picture above you mention the 9 pin connector seemingly for the main wiring harness. With the motor an unknow pin # connection. Is this accurate?
It appears to me that the new controller's Main harness is an 8 pin, with the Motor the 9...?

See my completed Magic Pie V5 rear hub motor E-Bike build HERE.



Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

11-03-2021, 12:59 AM #12

Hi Tommycat,

thanks a lot for the patient explanation!

> ***Please clarify, in your original controller's picture above you mention the 9 pin connector seemingly for the main wiring harness. With the motor an unknow pin # connection. Is this accurate?***
> ***It appears to me that the new controller's Main harness is an 8 pin, with the Motor the 9...?***

from the original controller there is a 9to5 Pins Juliet cable - the one that goes to the handle bar with display plug, breaks, etc, (its a 9to5 cause instead of the throttle theres a3 pins plug for the horn button + another plug for front light.), pict below

the other big plug from the orig controller is the typical motor plug with the big pins:

The Juliet cable I bought for the new controller is just a 4to1 cause I dont care about the horn.

unfortunately I dont have any wiring map :(

anyway I think I should be able to do the testing with the throttle you mentioned above,
I also read the other post and I was wondering:

- **can I do it using a 9V battery to connect the controller like you did in one of your pics or I have to use the original 48V batt from the bike?**
  that would make things easier for me...

one of your paragraph in the old post caught my attention:

> *Note: the controller 5 vdc regulator is not very powerful and is only rated to provide around 100mA or even less.(I've seen down to 40 mA)*
> ***Do not short out the supply wires.*** *It will survive a brief short, which I unfortunately can attest too. But probably not for an extended period, which I decline to verify...*
> *Insure correct battery power voltage to controller. Some controllers require an activation (key switch or button...) or "on" signal to it before energizing 5+vdc supply.*
> *Have a map?...Good use it*

My gut tells me this might be my case. As I mentioned this batt is always on, there is no switch button like the one you would have in a bafang system...
although of course I still have turn on the display with the power button, but I suspect the shitty cheap chinese original controller is tailor made to this peculiar thing of having a always on battery...

Anyway lemme know if I can do the test on a bench as I asked above and hopefully I'll come back with some results.
V.

Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

11-03-2021, 07:29 AM #13

Just noticed my last post is unapproved, I wonder why? can anyone see it?

Tommycat
Giga Member

Join Date: Oct 2017
Posts: 1256
Location: Northwestern Illinois

11-03-2021, 01:27 PM #14

I feel you might be missing a picture in the post above...?

> *Originally posted by **Valerio***
> *anyway I think I should be able to do the testing with the throttle you mentioned above.*
>
> *I also read the other post and I was wondering:*
>
> - ***can I do it using a 9V battery to connect the controller like you did in one of your pics or I have to use the original 48V batt from the bike?***
>   *that would make things easier for me...*

You can certainly ***bench test*** just your throttle, separate from your controller's power to see if it's operating properly. But the throttle's hall sensor minimum and maximum voltage input ratings are 2.7vdc to 6.5vdc. So using a 9 volt battery would not be advisable. (I don't ever remember using a 9vdc battery to test a throttle. A motor's hall sensors perhaps, as they are rated to 24vdc...)
A controller will typically put out a throttle supply voltage of 4.3vdc to 5vdc, that it gets from an energized 5vdc voltage regulator. So as I mention in the throttle thread... a 5vdc USB supply wall wort, 5vdc USB back-up type rechargeable battery, or 3-1.5vdc batteries in series (equaling 4.5vdc)would work well. Power usage is very low... As always your looking for .8vdc to 3.6vdc output on the signal leg.
Unless I'm missing the meaning of your questions... sorry.

But my whole point of the 5vdc discussion is to have you see ***if it is there*** available with the battery plugged in, and the display turned on. And to that end, the questions about the harnesses to get to the correct testing points that go to the throttle. And to see if perhaps something is shorting out that 5vdc regulated supply.

But I'm sure you know where the throttle goes, and can check for the voltage there...

"A quick check would be at the connector for the throttle. *TYPICALLY* 5vdc should be present between the controller's RED and BLACK throttle wires. "
and
"I would recommend disconnecting every thing except for the battery, and display. Turn on and check for error codes. And then for 5vdc output.
If all is well, plug in motor and re-check. Then throttle. Looking to see if at any time communication begins, and when it stops.
Look for and write down any error codes."

Note: A KT type display can be bypassed to energize the controller if desired by disconnecting the display and jumping the RED and BLUE wires, or terminals 1 and 2.

See my completed Magic Pie V5 rear hub motor E-Bike build HERE.

Valerio
Newbie

Join Date: Feb 2021
Posts: 16
Location: Singapore

11-04-2021, 03:13 AM #15

Here we go,

I tested the throttle cable with motor plug connected, it says 3.68V see below

I also tried with the motor plug disconnected and the value seems higher

Finally I tried with the throttle cable, another diff value

whats the verdict? Should I throw away everything and buy a toaster instead?